

Developing Containerised Solutions on the AWS Platform

Amazon offers a range of options for companies to elevate container orchestration, thus helping send products faster into the market. ECS, EKS, Fargate, Spinnaker and Amazon EKS Anywhere...the choice is varied.



AWS provides multiple choices for a containerised solution in the form of Elastic Container Service (ECS), Elastic Kubernetes Service (EKS) and the new EC2 Fargate. ECS and EKS are primarily meant for container orchestration — ECS is a pure AWS native cloud managed service and EKS is a Kubernetes platform-based native implementation.

When designing a Kubernetes container solution with Amazon EKS, the focus is on services and application groups, without placing significant emphasis on cost benefit and optimisation. The autoscaling feature handles automatic fan-in and fan-out of pod instances in EKS clusters.

Amazon EKS is chosen for managed Kubernetes control plane operations, with nodes based on Amazon EC2 instances which can scale faster in the cluster.

Amazon ECS and EKS

If you are familiar with Docker container handling, you can choose ECS-based architecture. If you are familiar with the Kubernetes framework, you can opt for an EKS-based architecture.

If you seek a cloud-agnostic solution, EKS is preferable, as it gives the advantage of using the underlying services offered by other cloud service providers as well (with less or no disruptions/changes in the level of

services). Also, if you are looking for a 'bin packing' solution with automatic resource management for the containers, EKS facilitates automated deployment, scaling of services, and self-management of containerised applications.

Amazon Fargate vs EKS

For those new to native container programming, Fargate is a sought-after solution for container orchestration. It is cost-effective as you pay only for computing time, not the underlying EC2 resources and instances.

Fargate is more advanced than ECS and EKS, providing a serverless computing engine to build and deploy containers on an ECS service. It alleviates concerns about configuring